

Coprimality is the vanishing of $[\Omega]$, but not the vanishing of ν : the cohomological locus of skew-brace decomposition and its orthogonal residue

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Abstract

The decomposition theory of finite skew braces has a cohomological shadow: an extension $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ of a skew brace by an ideal splits exactly when a certain bilinear second-cohomology class $[\Omega]$ vanishes. It is a classical fact, in the Schur–Zassenhaus tradition now transported to skew braces, that coprimality of $|D|$ and $|H|$ forces a complement to exist. We record that these two statements meet: on a finite but axiom-verified computational sweep (all skew braces on the additive groups of order at most 24), coprimality forces $[\Omega] = 0$ with zero exceptions across 111 instances, while $[\Omega] \neq 0$ occurs only in the non-coprime regime. We then correct a tempting misreading. One might expect coprimality to kill $[\Omega]$ by trivialising the *residue* ν , the outer action of H on D . It does not: ν survives, non-trivial in a majority of the decomposable coprime cases (with S_3 and A_4 as clean witnesses). Coprimality forces $[\Omega] = 0$ for a purely order-arithmetic reason, *orthogonally* to ν . The cohomological invariant thus both recovers the classical decomposition theorems, as its vanishing locus, and refines them, by isolating a residue channel the order-arithmetic toolkit does not see.

1 Introduction

The problem: when does a skew brace decompose? Skew braces are the algebraic backbone of set-theoretic solutions to the Yang–Baxter equation, and much of their structure theory is a theory of *decomposition*: given a skew brace G and an ideal $D \trianglelefteq G$, when does G split as an internal product of D with a complementary sub-skew-brace? A positive answer reduces the study of G to that of D and the quotient $H = G/D$ together with an action, exactly as a semidirect decomposition does for groups. Two lines of work bear on this. On the structural side, a Schur–Zassenhaus phenomenon has been established for skew braces: when $|D|$ and $|H|$ are coprime, a complement to D always exists [1].¹ On the cohomological side, the extension $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ is classified by a second cohomology, and splitting is detected by the vanishing of a distinguished class.

The cohomological shadow. In earlier work (peer-reviewed within our group) the present author isolated, for a skew-brace extension with D a proper non-trivial ideal, a *bilinear* obstruction class $[\Omega]$ with the property

$$[\Omega] = 0 \iff \text{the extension splits} \iff D \text{ has a sub-skew-brace complement.} \quad (1)$$

¹The general finite-skew-brace form of the statement — a complement for every ideal of coprime order or index — has been proved independently and contemporaneously by structure-theoretic methods (Crespo; Damele; Ferrara–Trombetti). Those sources are not yet at the citation floor used here, and we anchor the coprime-complement statement on the deep-read source [1], whose complement theorems cover the trivial-kernel case that our witnesses exercise.

The class $[\Omega]$ is the Rathee–Yadav extension class [1] in the trivial-kernel case; its bilinearity is genuine, and the equivalence (1) identifies the classical notion of decomposability with a cohomological vanishing. We take (1) as a premise throughout: it is the bridge that lets us ask a decomposition question as a cohomology question.

The gap. Given (1) and the coprime complement theorem, one sentence writes itself: *coprimality forces* $[\Omega] = 0$. But a sentence that writes itself deserves scrutiny on two counts. First, is it *informative* — is the obstruction genuinely realised off the coprime region (so the vanishing locus is a proper sublocus, not all of the ambient cohomology), and does a direct computation that trusts neither the premise (1) nor the cited theorem confirm the picture? Second, and more interestingly, *why*? A skew-brace extension carries a *residue* ν : the outer action of H on D , the image of the multiplicative action restricted to D in $\text{Aut}(D)$ modulo inner automorphisms. This residue is the channel through which the residue-dependent solvability obstruction o_ν of the cohomological extension theory operates. It is natural — it was, in fact, our own working hypothesis — to guess that coprimality kills $[\Omega]$ *through* ν : that coprime order trivialises the outer action, and with the action gone the obstruction has nowhere to live. If true, this would tie the splitting obstruction and the residue channel together at exactly the point where classical decomposition succeeds.

The result of this note. The hypothesis is half right and half wrong, and the wrong half is the interesting one. We verify computationally that coprimality does force $[\Omega] = 0$, without exception, on a finite but axiom-checked sweep; and we refute the mechanism. The residue ν is *not* trivialised by coprimality — it survives, and is generically non-trivial, in the very cases where G decomposes. Coprimality forces $[\Omega] = 0$ for an order-arithmetic (Schur–Zassenhaus) reason that is *orthogonal* to ν .

Observation 1. On the sweep of all skew braces whose additive group has order at most 24 (Section 4), for every proper non-trivial ideal D :

- (a) (*locus*) coprimality of $|D|$ and $|H|$ forces the extension to split, hence $[\Omega] = 0$, in all 111 coprime instances, with zero exceptions; and every split-failure (582 of them) occurs in the non-coprime regime;
- (b) (*orthogonality*) the residue ν is non-trivial in 70 of those 111 coprime splittings, and the H -action on D is non-trivial in 74 of them; the symmetric group S_3 and the alternating group A_4 are decomposable coprime witnesses with non-trivial residue.

This is a *computed* observation on a finite sweep, not a theorem: it confirms the locus statement and refutes the residue-trivialisation mechanism, but it does not prove the orthogonality in general. We are careful throughout to say which claims are established (the equivalence (1), peer-reviewed; the coprime complement theorem, cited) and which are computational (Observation 1).

Why it matters. The equivalence (1) is a first example of the cohomological machinery *explaining* a classical structural result: the decomposition theorems reappear as the description of a single vanishing locus. Observation 1 sharpens the picture by showing the machinery also *sees more* than the classical toolkit. The order-arithmetic proof of Schur–Zassenhaus is silent about ν ; the cohomological description makes ν a first-class invariant, one that persists as genuine structure even where decomposition succeeds. Recovering a classical theorem and isolating a channel it cannot see is exactly the kind of leverage a cohomological invariant should provide.

Outline. Section 2 fixes the language of skew-brace extensions, the class $[\Omega]$, and the residue ν . Section 3 records how (1) makes the coprime complement theorem a statement about the locus $\{[\Omega] = 0\}$. Section 4 describes the computational sweep and its axiom verification, and states the locus half of Observation 1. Section 5 states and refutes the residue-trivialisation hypothesis and gives the orthogonality half. Section 6 draws the moral and lists the open questions this note hands to a proof session.

2 Preliminaries

We recall only what the observation uses. All skew braces are finite.

Definition 2 (Skew brace). A *skew brace* is a set A with two group operations, $(A, +)$ (the *additive* group, not assumed abelian) and $(A, \circ)$ (the *multiplicative* group), sharing a unit 0 and satisfying the compatibility

$$a \circ (b + c) = (a \circ b) - a + (a \circ c) \quad (a, b, c \in A).$$

The associated λ -map $\lambda: (A, \circ) \rightarrow \text{Aut}(A, +)$, $\lambda_a(b) = -a + (a \circ b)$, records the additive action of the multiplicative group.

Definition 3 (Ideal and extension). An *ideal* of a skew brace A is a normal subgroup of $(A, \circ)$ that is also a normal, λ -invariant subgroup of $(A, +)$; the quotient A/D is then a skew brace. An *extension* of a skew brace H by an ideal D is a short exact sequence of skew braces

$$0 \longrightarrow D \longrightarrow G \longrightarrow H \longrightarrow 0,$$

i.e. $D \trianglelefteq G$ is an ideal with $G/D \cong H$. The extension *splits* if there is a sub-skew-brace $C \leq G$ with $D \cap C = 0$ and $D \circ C = G$; such a C is a *complement* to D , and G is then an internal (Zappa–Szép-type) product of D and C .

We write $|D|$ and $|H| = |G|/|D|$ for the orders; the extension is *coprime* if $\gcd(|D|, |H|) = 1$.

Definition 4 (The bilinear obstruction class). For an extension with D a proper non-trivial ideal there is a bilinear class $[\Omega]$ in second cohomology, agreeing with the Rathee–Yadav extension class $[(\beta, \tau)]$ in the trivial-kernel case [1], whose vanishing detects splitting, as in (1). We take (1) as given; it has been proved and peer-reviewed within our group.

Definition 5 (The residue ν). The *residue* of an extension is the outer action of H on D carried by the λ -map: concretely, the image of $\lambda|_D$ (the multiplicative action restricted to D) inside $\text{Aut}(D)$, read modulo inner automorphisms $\text{Inn}(D)$ and modulo the action of D on itself. We call the residue *trivial* when this image is trivial, i.e. when the outer H -action on D is inner; and we separately track whether the H -action on D is trivial at all (image of size 1 in $\text{Aut}(D)$). The residue is the datum on which the solvability obstruction o_ν depends, and it is the object whose fate under coprimality this note examines.

3 The $[\Omega] = 0$ locus recovers classical decomposition

The premise (1) is a translation, and translations are cheap; their value is in what they let one import. Here they let us import the Schur–Zassenhaus phenomenon.

Proposition 6 (Coprimalty lands in the locus). *Let $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ be a skew-brace extension with D a proper non-trivial ideal and $\gcd(|D|, |H|) = 1$. Then D has a complement, hence by (1), $[\Omega] = 0$.*

Proof. The existence of a complement in the coprime case is the skew-brace Schur–Zassenhaus statement: for coprime order there is a complementary sub-skew-brace [1]. Applying the equivalence (1) in the direction “complement $\Rightarrow [\Omega] = 0$ ” gives the conclusion. We claim no novelty for either input; the content of this note is what Sections 4–5 add on top of this proposition. $\square$

Proposition 6 says the classical decomposition theorem is the assertion that the coprime regime is contained in the cohomological locus $\{[\Omega] = 0\}$, and its contrapositive already settles that non-coprimality is *necessary* for $[\Omega] \neq 0$. Two questions remain that the proposition does not answer, and that motivate the rest of the note. First, is the containment *informative* — is the complementary locus $\{[\Omega] \neq 0\}$ actually populated, so that non-coprimality is necessary but far from sufficient, and does a direct computation, trusting neither the cited theorem nor the premise (1), confirm the whole picture? Second, through what mechanism does coprimality act — does it trivialise the residue ν , or bypass it? The first is answered by a census (Section 4), the second by inspecting the residue directly (Section 5).

4 The computational witness

Method. A Guarnieri–Vendramin λ -map constraint-propagation enumerator produced all skew braces on every additive group of order at most 24 in the list

$$\mathbb{Z}_4, \mathbb{Z}_2^2, \mathbb{Z}_6, S_3, \mathbb{Z}_8, \mathbb{Z}_2 \times \mathbb{Z}_4, \mathbb{Z}_2^3, D_4, Q_8, \mathbb{Z}_{12}, \mathbb{Z}_2 \times \mathbb{Z}_6, D_6, \text{Dic}_3, A_4, \mathbb{Z}_{15}, S_4.$$

The enumerator was independently *axiom-verified*: across all groups there were zero associativity or distributivity violations. For each proper non-trivial ideal D of each brace, splitting was decided *directly* — by searching for a complementary sub-skew-brace — rather than through any cohomological proxy. Deciding splitting directly is the point: it means the sweep tests the equivalence (1) against the ground truth of complements, not against a re-encoding of $[\Omega]$.

The census. Table 1 aggregates the outcome over all (brace, ideal) instances.

regime	instances	split-failures
coprime $\gcd(D , H) = 1$	111	0
non-coprime	1236	582

Table 1: Split-failures by regime, over all skew braces on additive groups of order ≤ 24 . Coprimality forces splitting with zero exceptions; every failure of decomposition occurs in the non-coprime regime.

Observation 7 (Locus half of Observation 1). On this sweep, all 111 coprime instances split, and all 582 split-failures are non-coprime. Equivalently, via (1): $[\Omega] = 0$ throughout the coprime regime, and $[\Omega] \neq 0$ occurs only when $\gcd(|D|, |H|) > 1$.

The non-coprime failures are not sporadic: they are witnessed already at $\mathbb{Z}_4$, at the order-8 group $W_4 = \mathbb{Z}_2 \times \mathbb{Z}_4$, at $\mathbb{Z}_2^3$, D_4 , Q_8 , and among the order-12 braces $\mathbb{Z}_{12}$, D_6 , Dic_3 . This is the region where $[\Omega] \neq 0$ lives, and it is exactly the region coprimality excludes.

Remark 8 (What the census is and is not). Observation 7 is a *computed witness* over a finite list of additive groups, not a proof, and it is not needed for the containment direction: that non-coprimality is *necessary* for $[\Omega] \neq 0$ is already Proposition 6 in general. The census adds two

things the proposition does not. First, it is an *independent* confirmation of the whole picture: splitting was decided by direct complement search, so the coprime 0/111 figure checks the cited Schur–Zassenhaus theorem and the premise (1) against ground truth rather than assuming them. Second, and substantively, it shows the obstruction is *non-vacuous*: $[\Omega] \neq 0$ genuinely occurs (582 times), so the locus $\{[\Omega] = 0\}$ is a proper sublocus and non-coprimality is necessary but far from sufficient — 654 of the 1236 non-coprime instances split regardless. What a finite census cannot do is characterise *which* non-coprime extensions obstruct; that is left open (Section 6).

5 The residue survives: orthogonality

We now come to the mechanism, and to the correction that is the reason for this note.

The hypothesis. Proposition 6 tells us *that* coprimality forces $[\Omega] = 0$; it is silent on *why*. A structurally attractive explanation — our own initial one — runs: coprimality trivialises the residue ν (Definition 5), and with the outer H -action on D gone, the bilinear obstruction, which is built from that action, must vanish. On this reading the splitting obstruction and the residue channel are the same phenomenon seen twice, coupled precisely where decomposition succeeds.

The refutation. The census records ν alongside splitting, and the record refutes the hypothesis.

Observation 9 (Orthogonality half of Observation 1). Of the 111 coprime splitting instances, 70 have *non-trivial* outer residue (the image of $\lambda|_D$ in $\text{Aut}(D)$ is not contained in $\text{Inn}(D)$), and 74 have a non-trivial H -action on D (image of size > 1 in $\text{Aut}(D)$). The residue is therefore non-trivial in a strict majority of the coprime decompositions.

Two clean, familiar witnesses make the point without a table:

- S_3 . Take $D = A_3 \cong \mathbb{Z}_3$ and $H \cong \mathbb{Z}_2$ acting by inversion. Then $\gcd(3, 2) = 1$, so the extension splits (indeed $S_3 = A_3 \rtimes \mathbb{Z}_2$), yet the outer action has image of order 2 in $\text{Aut}(\mathbb{Z}_3)$: the residue is non-trivial.
- A_4 . Take $D = V_4$ and $H \cong \mathbb{Z}_3$ acting by a 3-cycle. Then $\gcd(4, 3) = 1$, the extension splits ($A_4 = V_4 \rtimes \mathbb{Z}_3$), and the outer action has image of order 3 in $\text{Aut}(V_4) \cong S_3$: again non-trivial.

In both cases $[\Omega] = 0$ — the brace decomposes — while $\nu \neq 0$. So the vanishing of $[\Omega]$ under coprimality cannot be an effect of the vanishing of ν ; the two are decoupled.

Observation 10 (The corrected statement). Coprimality forces $[\Omega] = 0$ regardless of ν . The residue ν is orthogonal to the splitting obstruction: it survives, generically non-trivially, in the very cases where G decomposes. The mechanism by which coprimality kills $[\Omega]$ is order-arithmetic (Schur–Zassenhaus), not residue-trivialising.

Remark 11 (Reading the two counts). The 70 (outer residue non-trivial) and 74 (H -action non-trivial) differ because an H -action can be non-trivial yet inner — absorbed into $\text{Inn}(D)$ and so invisible to the outer residue. The gap of 4 measures exactly the coprime decompositions whose H -action is non-trivial but inner. Neither count is what the trivialisation hypothesis predicts, which is 0; both are large.

6 Conclusion

The moral. The bilinear class $[\Omega]$ does two things for skew-brace decomposition. It *recovers* the classical theorems: by (1), the coprime Schur–Zassenhaus complement theorem is the statement that the coprime regime lies in the vanishing locus $\{[\Omega] = 0\}$ (Proposition 6), and an independent census up to order 24 — deciding splitting by direct complement search — confirms the picture while showing the obstruction is genuinely realised ($[\Omega] \neq 0$ exactly 582 times, all non-coprime), so the locus is a proper sublocus (Observation 7). And it *refines* them: the residue ν , invisible to the order-arithmetic proof, is a genuine invariant that survives coprimality and is non-trivial in most decomposable coprime cases (Observations 9 and 10). A cohomological invariant that reproduces a classical result as a vanishing locus *and* sees a structural channel the classical toolkit cannot is doing real work.

Honest scope. The equivalence (1) is peer-reviewed; the coprime complement theorem is cited, not reproved; and the two central findings — the locus census (Observation 7) and the residue orthogonality (Observations 9–10) — are *computed* witnesses over a finite list of additive groups of order at most 24, not general theorems. “Residue trivial” means precisely the image of $\lambda|_D$ in $\text{Aut}(D)$ modulo $\text{Inn}(D)$ and modulo the D -action (Definition 5); a different bookkeeping of the residue could shift the counts but not the qualitative conclusion, since the S_3 and A_4 witnesses are unambiguous.

Open questions. Two are well-posed and hand naturally to a proof session.

Problem 12. Non-coprimality is necessary but not sufficient for $[\Omega] \neq 0$ (Proposition 6 gives necessity; the census finds 654 non-coprime splittings). Characterise, within the non-coprime regime, exactly which extensions obstruct — a sharp structural criterion for $[\Omega] \neq 0$ refining the order condition.

Problem 13. Prove that coprimality forces $[\Omega] = 0$ *orthogonally* to ν in general: exhibit the order-arithmetic vanishing of $[\Omega]$ as independent of the residue, so that Observation 10 becomes a theorem rather than a census. The expected shape is a Schur–Zassenhaus argument on $[\Omega]$ that never touches $\lambda|_D$.

Beyond these, the deep-reading of the recent structure-theoretic Schur–Zassenhaus papers for skew braces would let the general coprime-complement theorem be cited directly, in place of the trivial-kernel anchor used here; and the placement of $[\Omega]$ and ν against the homology leg of skew-brace theory [2] is pursued elsewhere.

References

- [1] M. Rathee and M. K. Yadav, *Skew brace extensions, second cohomology and complements*, arXiv:2601.12371.
- [2] M. Gran, T. Letourmy and L. Vendramin, *Hopf formulae for the homology of skew braces*, J. Pure Appl. Algebra **229** (2025), no. 11, 108085; arXiv:2409.18056.